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ENVIRONMENT

# Environmental Studies — Part 3

*Class 12*

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# Contents

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1. Ecosystems
2. Biodiversity and Conservation
3. Climate Change

## Chapter 1

# Ecosystems

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An ecosystem is a functional unit of nature where living organisms interact with each other and with their physical environment. Ecosystems consist of biotic components — producers, consumers and decomposers — and abiotic components such as light, temperature, water and soil. Energy flows through ecosystems via food chains and webs, with only about ten percent transferred between trophic levels. Biogeochemical cycles of carbon, nitrogen and phosphorus recycle essential elements. Understanding ecosystems is the basis of ecology and conservation.

## Chapter 2

# Biodiversity and Conservation

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Biodiversity refers to the variety of life at genetic, species and ecosystem levels. India, a megadiverse country, harbors a rich variety of flora and fauna. Biodiversity is threatened by habitat loss, overexploitation, pollution, invasive species and climate change. Conservation strategies include in-situ methods such as national parks and sanctuaries, and ex-situ methods such as zoos and seed banks. The Convention on Biological Diversity and national laws aim to protect and sustainably use biodiversity for future generations.

## Chapter 3

# Climate Change

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Climate change refers to long-term shifts in temperature and weather patterns, driven largely by human activities that release greenhouse gases such as carbon dioxide and methane. The Intergovernmental Panel on Climate Change documents rising global temperatures, melting glaciers, sea-level rise and increased frequency of extreme events. Mitigation requires reducing emissions through renewable energy, energy efficiency and afforestation. Adaptation involves building resilience in agriculture, infrastructure and communities. Climate justice emphasizes equitable burden-sharing between developed and developing nations.